National Land Acquisition & Management System (NLAMS)

Smart India Hackathon (SIH) 2026

Web3 & GIS Enabled End-to-End Monitoring Portal Report

PROJECT METADATA

Problem Statement:	**Real-Time National Land Acquisition & Management System**
Focus Areas:	**Web3 Blockchain, GIS Parcel Mapping, LARR Workflow Router**
Prototype Version:	**v1.0 (Fully Responsive React / Vite)**
Compiled Date:	**August 2026**

1. How to Share the Prototype with Your Team

To allow your hackathon leader and other team members to preview, test, and interact with the frontend, you can choose from these sharing options:

Option A: Free Cloud Deployment (Highly Recommended)

Deploying the frontend to a cloud platform gives you a public link (e.g., project.vercel.app) that anyone can access from their phones or laptops instantly.

- **Vercel (Easiest):** Install the Vercel CLI via npm (npm install -g vercel), run vercel in the frontend folder, and follow the 2-step setup to get a live URL.
- **Netlify:** Drag and drop the generated frontend/dist directory directly onto Netlify Drop (app.netlify.com) for a 10-second instant deployment.
- **GitHub Pages:** Create a GitHub repository, push the codebase, and configure GitHub Pages to build from the main branch using Vite actions.

Option B: Temporary Public Tunnel (For Live Demos)

If you want to share the website directly from your local machine while you are coding, you can tunnel your local port to the internet:

- **Localtunnel:** Run npx localtunnel --port 5173 while your development server is active. It will generate a public link you can share with your team.
- **Ngrok:** Run ngrok http 5173 (requires free ngrok account signup) to establish a secure public HTTPS gateway directly to your localhost.

Option C: Sharing over Local Network (WiFi)

If your team is in the same room connected to the same WiFi network, they can open the site using your computer's local IP address:

- **Vite Host Flag:** Start your server with npm run dev -- --host. Vite will output your local IP (e.g. http://192.168.1.10:5173). Anyone on the same network can load that link.

2. Overall Website Layout & Page Features

The NLAMS system translates the LARR (Land Acquisition Act, 2013) guidelines into a unified digital pipeline. The pages serve the following purposes:

Home / Landing Portal

Acts as the front gate. Summarizes the system goals, features visual key pillars (decentralized audits, GIS map visualization, smart escrows, and R&R support), and defines the stakeholder workflow sequence for LARR compliance.

Executive Analytics & Monitoring (Dashboard)

Designed for central and state decision-makers. Features:

- **Dynamic KPIs:** Tracks total notified vs acquired land area, compensation disbursement ratios, and average resettlement rates.
- **GIS Visualizer:** Renders land parcel coordinates as interactive map polygons. Colors update dynamically based on stage.
- **AI Predictive Analytics:** Provides risk alerts, completion forecasts, and bottleneck analysis based on regional parcel fragmentation and current step latency.
- **MIS Export:** An active button compiling project metrics into a downloadable CSV spreadsheet on the fly.

Proposals & Automated Workflows

The operational pipeline. Central ministries can register new proposals. State, district, and surveyor profiles can switch roles in the header and verify/approve subsequent milestones (GIS validation, Sec 11 legal notices, Award finalizing, and final possession handover).

Web3 Trust & Compensation Portal

The decentralised backend simulation. Integrates a mock Metamask connection to disburse escrow funds. Logs every action (payouts, titles, geo-fences) in a visual scrolling blockchain ledger. Includes a digital deed upload widget that computes file SHA-256 hashes to verify authenticity.

Field Surveyor Node

Mobile-responsive field interface. Simulates GPS sensor triangulation to capture boundary points. Records valuation attributes (soil fertility, structure types) and uploads site photos.

3. Tech Stack Breakdown

The application is developed using standard industry-level technologies suited for fast hot-reloading and polished UI output under pressure:

- **Frontend Framework:** React.js built on Vite (Vite 8, React 19) for immediate loading times and HMR (Hot Module Replacement).
- **Styling Stack:** Tailwind CSS v4 for fully custom responsive components, clean government-themed color palettes, and layouts.
- **GIS Map Rendering:** React Leaflet & Leaflet.js utilizing OpenStreetMap tile layers to draw and render coordinate boundaries (polygons).
- **MIS Visual Charts:** Recharts package using SVG elements for responsive financial and progress bar graphs.
- **Icons:** Lucide React icons for clean, minimalistic government and technical emblems.
- **Web3 Simulation:** Context-based mock ledger engine simulating SHA-256 hashes, transaction block creation, and crypto signatures.

4. Path to Production: Making the Project Real

To elevate this frontend prototype into a production-grade system for the final hackathon round, the following back-end integrations are required:

A. Smart Contract Deployments (Web3)

Replace the mockup ledger with actual Solidity Smart Contracts deployed to an Ethereum Layer 2 (e.g. Arbitrum, Optimism) or Polygon. Use contracts to tokenize land titles as Soulbound NFTs (non-transferable title deeds) and lock compensation in Escrow wallets that auto-release on PFMS verification.

B. Real API Integrations

Connect actual government databases to feed data into the portal:

- **BHOOMI / State Registries:** Connect REST APIs to fetch official land ownership details, preventing double-selling disputes.
- **BhuNaksha:** Import XML/GeoJSON cadastral maps to verify physical land shapes on the map.
- **PFMS Gateway:** Integrate the Public Financial Management System to trigger real-time bank transfers.

C. Geospatial Database

Use a database like PostgreSQL with the PostGIS extension. This allows you to run spatial SQL queries (e.g. detect if highway path intersects with protected forest reserves or existing private buildings).

D. Official Authentication

Implement Single Sign-On (SSO) using India's Digilocker or Aadhaar API, verifying the identities of surveyors, district magistrates, and farmers.